

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 35-2021: A 50-Year-Old Woman with Pain in the Left Upper Quadrant and Hypoxemia

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PRESENTATION OF CASE

Dr. Mubeen A. Shakir (Medicine): A 50-year-old woman was admitted to this hospital because of pain in the left upper quadrant and hypoxemia.

The patient had been in her usual state of health until 1 year before this admission, when fatigue and weight loss developed and she had abnormal results on liver-function tests after undergoing cholecystectomy for presumed acalculous cholecystitis. Computed tomography (CT) of the abdomen, which was performed as part of the evaluation to determine the cause of the abnormal liver-function test results, revealed hepatosplenomegaly (Fig. 1A).

Seven months before this admission, a submandibular mass and swelling developed; 1 month later, examination of a biopsy specimen of the mass revealed granulomatous sialadenitis. A liver biopsy was performed to investigate the patient's abdominal distention, poor appetite, and persistently abnormal liver-function test results, which were suggestive of cholestasis. The liver-biopsy specimen showed evidence of granulomatous hepatitis; portal and periportal nonnecrotizing granulomatous inflammation that was associated with multinucleated giant cells, lymphocytes, and some plasma cells; and stage 1 to 2 (portal and periportal) fibrosis. Special stains for organisms (acid-fast and Grocott methenamine–silver stains) were negative.

CT of the chest revealed minimal bibasilar atelectasis and a trace left pleural effusion. An interferon- γ release assay for *Mycobacterium tuberculosis* was negative. Tests for histoplasma species, blastomyces species, cryptococcus species, coccidioides species, the human immunodeficiency virus (HIV), and hepatitis A, B, and C viruses were all negative, as was testing for antinuclear antibodies, antimitochondrial antibodies, and anti-smooth muscle antibodies. A diagnosis of sarcoidosis was made on the basis of the biopsy results of the submandibular gland and liver and the presence of hypercalcemia and an elevated angiotensin-converting-enzyme level.

Four months before this admission, prednisone therapy was initiated at a dose of 40 mg per day; subsequently, there was a marked reduction in the patient's abdominal distention and marked improvement in her appetite. Liver-function test

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N Engl J Med 2021;385:1995-2001.

DOI: 10.1056/NEJMcpc2107356

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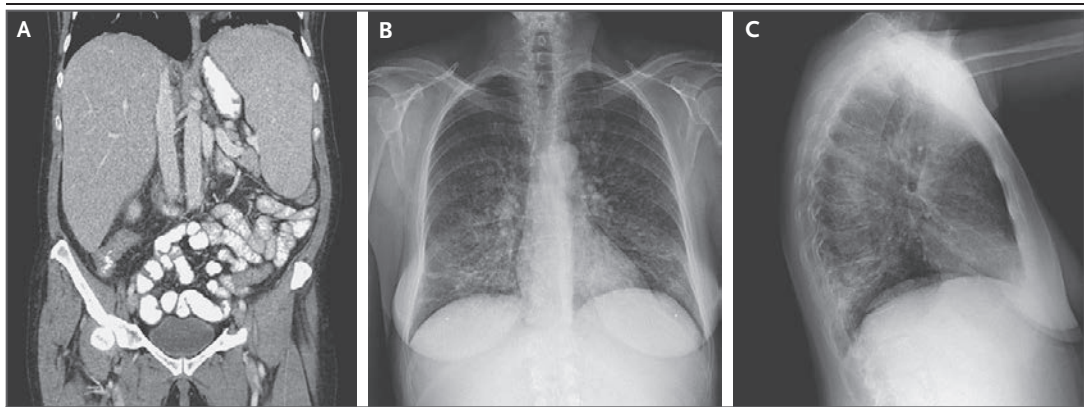


Figure 1. Imaging Studies Obtained before Admission.

A CT scan of the abdomen obtained 8 months before admission (Panel A) shows marked hepatosplenomegaly. A chest radiograph obtained 1 month before admission (Panels B and C) shows bilateral, perihilar, diffuse ground-glass opacities with additional patchy opacities at the lung bases and no evidence of pleural effusion.

results also improved. Methotrexate therapy was started at a dose of 15 mg per week, with plans to taper the dose of prednisone.

Three months before this admission, the patient stopped prednisone therapy because of irritability and insomnia. One month later, routine monitoring of the patient's liver function revealed worsening liver-function test results in a cholestatic pattern. This finding was attributed to undertreatment of sarcoidosis, but the patient declined to restart prednisone; the dose of methotrexate was increased to 25 mg per week.

During the next month, the patient noted fatigue, slowly progressive dyspnea on exertion, and cough, without fever or chills. She was evaluated by her primary care physician 1 month before this admission, and a chest radiograph showed bilateral, perihilar, diffuse, symmetric ground-glass opacities with scattered patchy opacities at the lung bases (Fig. 1B and 1C). Levofloxacin was administered.

Three weeks before this admission, CT of the abdomen and pelvis revealed worsening hepatosplenomegaly. Prednisone therapy was started at a dose of 20 mg per day.

On the morning of admission, the patient was awakened from sleep by severe, sharp pain in the left upper quadrant. The pain was not worse with eating, and she reported no nausea, vomiting, or diarrhea. Because of persistent pain throughout the day, she presented to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported progressive dyspnea on exertion and being unable to climb stairs in her home. There was no chest pain, edema in the legs, weight gain, or orthopnea. She had a cough that was productive of clear sputum but had no fevers or chills.

The patient had a history of preeclampsia, *Helicobacter pylori* infection, cholecystectomy, and papillary thyroid cancer for which she had undergone hemithyroidectomy 9 years before this admission. In addition to methotrexate and prednisone, her medications included omeprazole and trazodone. Her father had hypertension and hyperlipidemia, and one son had ulcerative colitis. Both maternal grandparents had had tuberculosis, but the patient had not had contact with them; her mother had latent tuberculosis. The patient had no known drug allergies. She worked as an administrative assistant and lived with her husband and children in a suburb of Boston. She had never smoked tobacco, did not use illicit drugs, and previously drank one glass of wine per week before her diagnosis of sarcoidosis. She had not traveled outside the United States in the past 5 years.

On examination, the temperature was 36.7°C, the blood pressure 112/56 mm Hg, the pulse 128 beats per minute, the respiratory rate 22 breaths per minute, and the oxygen saturation 85% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 21.

The patient was awake and alert but appeared fatigued. The first and second heart sounds (S₁ and S₂) were normal, without murmurs; the jugular venous pressure, which was measured from the approximate midpoint of the right atrium, was 6 cm of water. Crackles were present in the middle and lower lung fields bilaterally. Normal bowel sounds were present, and the abdomen was soft and nondistended, with tenderness in the left upper quadrant. The liver edge was palpable 3 cm below the right costal margin; the spleen tip was also palpable. There was no edema in the legs. Supplemental oxygen was administered through a nasal cannula at a rate of 3 liters per minute, and the oxygen saturation increased to 93%.

The blood levels of electrolytes, glucose, and lactate were normal, as were the results of kidney-function tests. The level of alanine aminotransferase was 38 IU per liter (reference range, 7 to 33), and the levels of aspartate aminotransferase and total bilirubin were normal. The blood level of alkaline phosphatase was 268 IU per liter (reference range, 30 to 100). *Streptococcus pneumoniae* and *Legionella pneumophila* antigens were not detected in the urine. The blood level of 1,3- β -D-glucan was less than 31 pg per milliliter (reference value, <60); other laboratory test results are shown in Table 1.

Dr. Victorine V. Muse: CT of the abdomen and pelvis (Fig. 2A and 2B), performed after the administration of intravenous contrast material, revealed hepatosplenomegaly with a hypodense wedge-shaped region in the midpole of the spleen. A portable anteroposterior chest radiograph (Fig. 2C) showed progressive coalescing consolidations in the middle and lower lobes with diffuse ground-glass opacities. CT of the chest (Fig. 2D), performed after the administration of intravenous contrast material, revealed new mediastinal and hilar lymphadenopathy and confirmed the presence of multifocal confluent areas of airspace consolidation, with a background of diffuse ground-glass opacities. There was no evidence of pulmonary embolism or pleural effusions.

Dr. Shakir: The patient was admitted to the hospital. Treatment with prednisone, methotrexate, and omeprazole was stopped, and trazodone was continued. On hospital day 2, the temperature rose to 39.1°C, the respiratory rate increased

Table 1. Laboratory Data.

Variable	Reference Range, Adults*	On Admission, This Hospital
Hematocrit (%)	36–46	35.3
Hemoglobin (g/dl)	12–16	11.1
White-cell count (per μ l)	4500–11,000	16,890
Differential count (per μ l)		
Neutrophils	1800–7700	8280
Lymphocytes	1000–4800	7430
Monocytes	200–1200	840
Eosinophils	0–900	170
Platelet count (per μ l)	150,000–400,000	183,000
Mean corpuscular volume (fl)	80–100	90.3
Prothrombin time (sec)	11.5–14.5	14.8
Prothrombin-time international normalized ratio	0.9–1.1	1.2
D-dimer (ng/ml)	<500	2897
Lactate dehydrogenase (U/liter)	110–210	338
C-reactive protein (mg/liter)	<8	94.6

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

to 32 breaths per minute, and the oxygen saturation decreased to 89% while she was receiving supplemental oxygen through a nasal cannula at a rate of 3 liters per minute. Supplemental oxygen was increased to a rate of 5 liters per minute, and the oxygen saturation increased to 93%. Cultures of the blood were obtained.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Margaret M. Chapman: This 50-year-old woman who has been receiving immunosuppressive medication for the treatment of sarcoidosis presents with subacute, accelerating pulmonary symptoms and is found to have tachypnea, hypoxemia, and fever with progressive bilateral consolidations on chest imaging. Evaluation of the clinical timeline, current severity of illness, and imaging findings allows me to create a problem representation that informs my diagnostic thought process. I will use these elements as a filter through which I will develop a differential diagnosis. To do so,

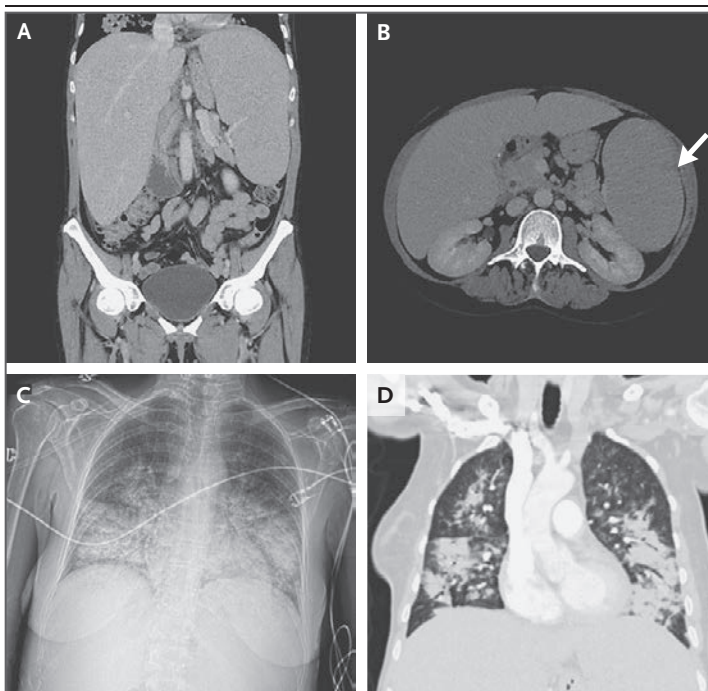


Figure 2. Imaging Studies Obtained on Admission.

CT scans of the abdomen and pelvis obtained on admission (Panels A and B) show persistent hepatosplenomegaly with a hypodense lesion in the spleen (Panel B, arrow). A portable anteroposterior chest radiograph (Panel C) shows progressive coalescing consolidations in the middle and lower lobes with diffuse ground-glass opacities. A CT scan of the chest (Panel D) confirms these findings and also shows new mediastinal and hilar lymphadenopathy.

there are three discrete questions to address: Does the patient really have sarcoidosis? Could her current presentation be a manifestation or complication of pulmonary sarcoidosis? Could her illness be due to a complication from the treatment of sarcoidosis?

SARCOIDOSIS

We often assume that a patient's medical history is correct, but in this patient who presents with a progressive illness of unknown cause, it is worth reevaluating the recent diagnosis of sarcoidosis. Sarcoidosis is a challenging diagnosis to make and can have myriad clinical manifestations. It is theoretically possible that this patient's symptoms that began 7 months before this admission were due to something other than sarcoidosis and that her current presentation is a manifestation of that other underlying illness. Other than sarcoidosis, what could ex-

plain the combination of hepatosplenomegaly, abnormal results on liver-function tests, and a biopsy showing hepatic granulomas?

On the basis of previous negative testing, tuberculosis; fungal diseases such as cryptococcosis, blastomycosis, and histoplasmosis; and infection with HIV, hepatitis A virus, hepatitis B virus, hepatitis C virus, and coccidioidomycosis are all unlikely diagnoses. An autoimmune process such as primary biliary cirrhosis is unlikely on the basis of negative serologic testing. The patient has not been receiving medications that can lead to hepatic granulomas, such as allopurinol, quinidine, sulfa-containing agents, or checkpoint inhibitors, and she has not received the bacille Calmette–Guérin vaccine. Evidence of cancer would probably have been apparent on her imaging studies and biopsy. I suspect that sarcoidosis is indeed the most likely explanation for her initial presentation 7 months before this admission.

MANIFESTATIONS AND COMPLICATIONS OF PULMONARY SARCOIDOSIS

The patient's treatment with glucocorticoids has been intermittent, which raises the possibility that she has sarcoidosis that has been inadequately treated, along with new pulmonary manifestations. Although sarcoidosis commonly affects the lungs, pulmonary sarcoidosis would not match the patient's presentation in terms of the three clinical filters through which I am viewing this case: time course, clinical severity, and imaging findings. Pulmonary sarcoidosis often manifests indolently, and patients may be asymptomatic for a long period of time, which is unlike this patient's accelerating course. The common symptoms of cough and dyspnea do not match this patient's clinical presentation of hypoxemia and high fever. Finally, pulmonary sarcoidosis can be manifested by a variety of radiographic patterns, but none are consistent with this patient's bilateral patchy and consolidative infiltrates that are present predominantly in the lower lobes.

Other possible causes of this patient's current presentation are potential complications of sarcoidosis, including infection, venous thromboembolism, and pulmonary hypertension. Secondary bacterial infections, such as those caused by streptococcus, staphylococcus, or atypical bacte-

ria, are possible, but the lack of improvement in her condition after a course of broad-spectrum antibiotic agents makes a bacterial cause unlikely. A fungal infection, such as aspergillosis, could be considered, but the radiographic findings in this case are not consistent with such an infection. Venous thromboembolism can be a complication of pulmonary sarcoidosis, but I would not expect a high fever or radiographic findings such as those observed in this patient to be associated with venous thromboembolism. In addition, although pulmonary hypertension can develop in patients with sarcoidosis, the subacute, accelerating time course seen in this patient would be unusual. Given that none of these possibilities are consistent with the time course of her illness, her clinical symptoms, or her radiographic findings, both progression of sarcoidosis and complication of sarcoidosis can be ruled out.

COMPLICATIONS FROM THE TREATMENT OF SARCOIDOSIS

When I consider the possibility of complications from the patient's recent medication use related to the treatment of her sarcoidosis, I separate the complications into two categories: medication effect and infections due to immunosuppression. Although prednisone has multiple effects on the body, it does not typically affect the lungs, so I will remove it from consideration. Methotrexate can have two types of effects on the lungs: lymphoproliferative and inflammatory. A lymphoproliferative effect on the lungs would produce nodules and masses. However, such a manifestation would not be consistent with the radiographic pattern seen in this patient, nor would it fit with the accelerating time course of her illness; therefore, I think that a lymphoproliferative effect from methotrexate is unlikely to explain her presentation.

However, a direct inflammatory effect from methotrexate is worth consideration. This effect would manifest as hypersensitivity pneumonitis, a disorder that often occurs weeks to months after initiation of methotrexate. Patients with hypersensitivity pneumonitis from methotrexate may present with cough, dyspnea, hypoxemia, and fever, as this patient does, and chest radiography often reveals an interstitial infiltrate predominantly in the lower lung fields.¹ In summary,

a direct inflammatory effect of methotrexate is consistent with this patient's time course, clinical severity, and radiographic findings and remains a possible diagnosis in this case.

Could an infection occurring in the context of immunosuppressive medication use explain this patient's presentation? Assessment of the net state of immunosuppression in a person receiving immunosuppressive medications is often difficult. However, I consider this patient to be moderately immunosuppressed on the basis of her receipt of moderate doses of prednisone and methotrexate. She has a cumulative prednisone exposure of almost 2 grams. She is therefore at risk for both opportunistic infections as well as more common infections.² With a focus on infections that could cause pneumonia and make a person as clinically ill as this patient, I considered the possibility of infection with *Staphylococcus aureus*, *S. pneumoniae*, community-acquired atypical bacteria (e.g., mycoplasma, legionella, or chlamydophila), nocardia, *Pneumocystis jirovecii*, or cryptococcus species. Of these, only infection with nocardia, *P. jirovecii*, and cryptococcus species could follow the time course of a subacute, accelerating illness; infection with the other pathogens would have a much faster clinical time course. Only *P. jirovecii* pneumonia is consistent with the radiographic pattern seen on this patient's imaging studies, which makes it the most likely infectious cause of this patient's illness.

I am left with a differential diagnosis of methotrexate-induced hypersensitivity pneumonitis and *P. jirovecii* pneumonia. *P. jirovecii* pneumonia appears to be the most likely diagnosis, since this condition is more common than methotrexate-induced hypersensitivity pneumonitis. Unfortunately, the totality of the data provided in this case does not perfectly support that diagnosis. This patient's serum assay for 1,3- β -D-glucan, a test with at least 90% sensitivity to diagnose *P. jirovecii* pneumonia, is negative.³ However, the pretest probability of *P. jirovecii* pneumonia in this patient is very high, and all the remaining data for this patient are consistent with this diagnosis. Thus, a negative serum 1,3- β -D-glucan test is not compelling enough as evidence to convince me that *P. jirovecii* pneumonia is not the correct diagnosis. I suspect that the diagnostic test that was performed in this case

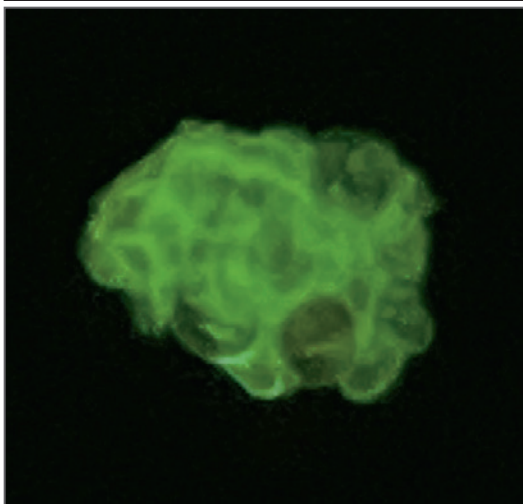


Figure 3. Bronchoalveolar-Lavage Specimen.

Microscopic examination of *Pneumocystis jirovecii*-specific fluorescent antibody staining of the patient's bronchoalveolar-lavage fluid reveals the diagnostic morphologic features of apple-green-stained clusters of *P. jirovecii* cysts. The cysts appear as darker, hollowed-out regions.

was direct fluorescent antibody staining of an induced-sputum specimen to test for *P. jirovecii*.

DR. MARGARET M. CHAPMAN'S DIAGNOSIS

Pneumocystis jirovecii pneumonia.

DIAGNOSTIC TESTING

Dr. Melis N. Anahtar: The diagnostic test that was performed was a direct fluorescent antibody test that was specific for *P. jirovecii*; microscopic examination of the patient's bronchoalveolar-lavage fluid revealed rare aggregates of *P. jirovecii* cysts (Fig. 3). *P. jirovecii* is a fungus with a complex life cycle and cannot be grown under standard culture conditions. It is identified either with the use of direct microscopic visualization, such as with the highly specific direct immunofluorescence staining used in this case, or with a polymerase-chain-reaction (PCR) test. As compared with direct visualization, a PCR test is more sensitive but less specific owing to its greater ability to detect asymptomatic carriage.⁴

In addition to the direct-detection methods described above, a blood test for 1,3- β -D-glucan (a cell-wall polysaccharide found in most fungi,

including *P. jirovecii*) is often used as a noninvasive diagnostic test for *P. jirovecii* pneumonia. The test has an overall sensitivity of 91% and an overall specificity of 79%.⁵ However, this patient's 1,3- β -D-glucan blood test was negative, at less than 31 pg per milliliter (reference value, <60), despite diagnostic confirmation of *P. jirovecii* pneumonia. This finding reflects the fact that the 1,3- β -D-glucan blood test is less sensitive in persons without HIV infection than in those with HIV infection or acquired immunodeficiency syndrome (AIDS), probably owing to a lower fungal burden, and must be interpreted within the context of the pretest probability of *P. jirovecii* pneumonia.⁵⁻⁷

MICROBIOLOGIC DIAGNOSIS

Pneumocystis jirovecii pneumonia.

DISCUSSION OF MANAGEMENT

Dr. James E. Mojica: *P. jirovecii* is an atypical fungus that scavenges host cholesterol for its membranes rather than synthesizing ergosterol⁸; therefore, treatment with azole antifungal agents or amphotericin is not effective. Caspofungin and micafungin inhibit 1,3- β -D-glucan synthetases and are effective against the cystic form of *P. jirovecii*. However, these agents are ineffective against the trophic form of *P. jirovecii* and thus do not fully eradicate the infection. Trimethoprim-sulfamethoxazole is the cornerstone of pneumocystis pneumonia treatment. This antibiotic combination disrupts folate synthesis crucial to the growth and reproduction of the organism by blocking two enzymes that are not expressed in humans (dihydrofolate reductase and dihydropyrimidine synthase). The intravenous route of administration of trimethoprim-sulfamethoxazole is standard in critically ill patients with *P. jirovecii* pneumonia, but oral therapy is acceptable in patients with mild disease.

The adjunctive use of glucocorticoids in patients with *P. jirovecii* pneumonia has been evaluated in persons with *P. jirovecii* pneumonia in the context of AIDS. The observation that the condition of these patients worsened after initiation of antipneumocystis therapy prompted the addition of glucocorticoid therapy in an attempt to reduce inflammation and acute lung injury. Several small randomized trials (involving 38 to

250 patients) that were conducted throughout the United States and Europe evaluated the role of glucocorticoids given early in the course of *P. jirovecii* pneumonia in patients with AIDS. When used early, glucocorticoids were found to reduce the risk of progression of hypoxemia, respiratory failure, and even death in patients who had a partial pressure of arterial oxygen of less than 70 mm Hg or an alveolar–arterial gradient above 35 mm Hg.⁹ It is important to note that randomized trials involving persons without HIV infection, such as this patient, are lacking.

This patient was treated first with intravenous trimethoprim–sulfamethoxazole and was later transitioned to oral therapy to complete a 3-week course. She also received adjunctive prednisone therapy, given the degree of her hypoxemia. She was discharged home with instructions to use supplemental oxygen during exertion, as needed. At a follow-up visit with a rheumatologist, her shortness of breath had resolved, and the oxygen saturation was 98% while she was breathing ambient air. After she completed the prednisone therapy that she had been receiving for treatment of *P. jirovecii* pneumonia, methotrexate was restarted at a dose of 20 mg per week for treatment of sarcoidosis. Follow-up CT of the chest revealed clearance of the ground-glass opacities.

A physician: Should this patient have been receiving prophylaxis against *P. jirovecii* pneumonia?

Dr. Mojica: For some patients, such as those who have AIDS and CD4+ T-cell counts of less than 200 per microliter, use of prophylaxis against *P. jirovecii* pneumonia is standard practice. Among patients without HIV infection, guidelines exist

for those with hematologic cancers, for those with solid cancers who are being treated with cytotoxic chemotherapy, and for those undergoing organ or stem-cell transplantation. For patients who are immunosuppressed because of inflammatory conditions, such as this patient, it is a more nuanced decision. I typically recommend prophylaxis against *P. jirovecii* pneumonia for patients who are receiving prednisone at a dose of 20 mg or higher per day for more than 4 weeks. There is no consensus regarding the routine use of prophylaxis against *P. jirovecii* pneumonia for patients being treated with methotrexate. Historically, the incidence of *P. jirovecii* pneumonia among patients treated with methotrexate monotherapy has been quite low, and many physicians do not prescribe prophylaxis against *P. jirovecii* pneumonia in such a scenario. However, the American Thoracic Society guidelines recommend prophylaxis against *P. jirovecii* pneumonia for patients treated with methotrexate. In this patient who was treated with both prednisone and methotrexate, I would prescribe prophylaxis against *P. jirovecii* pneumonia while monitoring for synergistic bone marrow suppression in the context of concurrent methotrexate and trimethoprim–sulfamethoxazole therapy.

FINAL DIAGNOSIS

Pneumocystis jirovecii pneumonia.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. John Branda for his assistance with the pathology portions of the case presentation.

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